

LiDAR–Camera Fusion for Road Segmentation Using a Modified Encoder–Decoder Network

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Abstract—Reliable road segmentation is a fundamental perception task for autonomous driving, yet camera-only methods can be sensitive to illumination, appearance variation, and adverse environmental conditions. This work investigates whether complementary LiDAR geometry can improve binary road segmentation when fused with RGB imagery. A SegNet-inspired encoder–decoder network is modified for road segmentation and evaluated under three sensor configurations: RGB only, early LiDAR–camera fusion, and late feature fusion. KITTI camera images and Velodyne point clouds are geometrically aligned using the provided calibration parameters, and the projected LiDAR measurements are converted into image-aligned depth representations. The models are trained using TensorFlow/Keras with GPU acceleration. The modified RGB-only network improves the reported MaxF score from 0.8142 for the baseline SegNet implementation to 0.9010. Incorporating LiDAR further improves MaxF to 0.9153 with early fusion and 0.9364 with late fusion. Among the evaluated configurations, late fusion achieves the best overall performance, with an accuracy of 0.7958 and an Intersection over Union (IoU) of 0.4898, at the cost of increased training time due to the use of separate encoder branches. These results demonstrate that LiDAR-derived geometric information can complement RGB appearance cues for road segmentation and that the location of multimodal fusion within the network materially affects performance.

Index Terms—LiDAR–camera fusion, semantic segmentation, road detection, SegNet, multimodal perception, autonomous driving, sensor fusion

I. INTRODUCTION

Reliable perception of the road surface is essential for autonomous vehicles because downstream planning and control depend on an accurate understanding of drivable space. Camera-based semantic segmentation has become a standard approach for dense scene understanding, supported by advances in convolutional neural networks (CNNs) such as VGG [1], fully convolutional networks (FCNs) [2], and encoder–decoder architectures such as SegNet [3]. However, RGB imagery alone can be sensitive to illumination changes, overexposure, low contrast, and adverse weather.

Complementary sensing modalities can reduce some of these limitations. LiDAR provides direct geometric measurements and accurate range information that are less dependent on scene appearance. Prior work has therefore explored multimodal perception using LiDAR [4], radar [5], and thermal imagery [6]. LiDAR–camera fusion is particularly attractive

because it combines dense visual appearance with sparse but metrically meaningful 3D structure.

This work studies LiDAR–camera fusion for binary road segmentation using the KITTI road dataset. LiDAR points are projected into the camera image plane using the supplied calibration parameters and converted into an image-aligned depth representation. A modified SegNet-inspired encoder–decoder is then evaluated under three configurations: RGB only, early fusion, and late feature fusion. The main contributions of this project are:

- an image-aligned LiDAR preprocessing pipeline based on KITTI calibration data;
- a modified encoder–decoder network for binary road segmentation;
- implementation of early and late LiDAR–camera fusion strategies; and
- a quantitative comparison of segmentation performance and training cost across the three sensor configurations.

II. RELATED WORK

A. Semantic Segmentation

FCNs demonstrated that image-classification CNNs can be converted into dense prediction networks by replacing fully connected layers with convolutional layers [2]. Skip connections between coarse semantic features and higher-resolution intermediate features improved localization quality in FCN-16s and FCN-8s.

SegNet introduced an efficient encoder–decoder structure for pixel-wise semantic segmentation [3]. Its encoder follows the convolutional structure of VGG16, while the decoder reuses max-pooling indices to recover spatial resolution. This design reduces decoder memory requirements and provides sharper boundary reconstruction than approaches that retain all encoder feature maps.

Real-time segmentation methods such as ICNet reduce computational cost by processing multiple image resolutions and fusing coarse semantic context with higher-resolution detail [7]. PSPNet similarly improves scene understanding by aggregating contextual information at multiple spatial scales [8]. These methods motivate the use of context aggregation and multi-scale feature processing for road-scene segmentation.

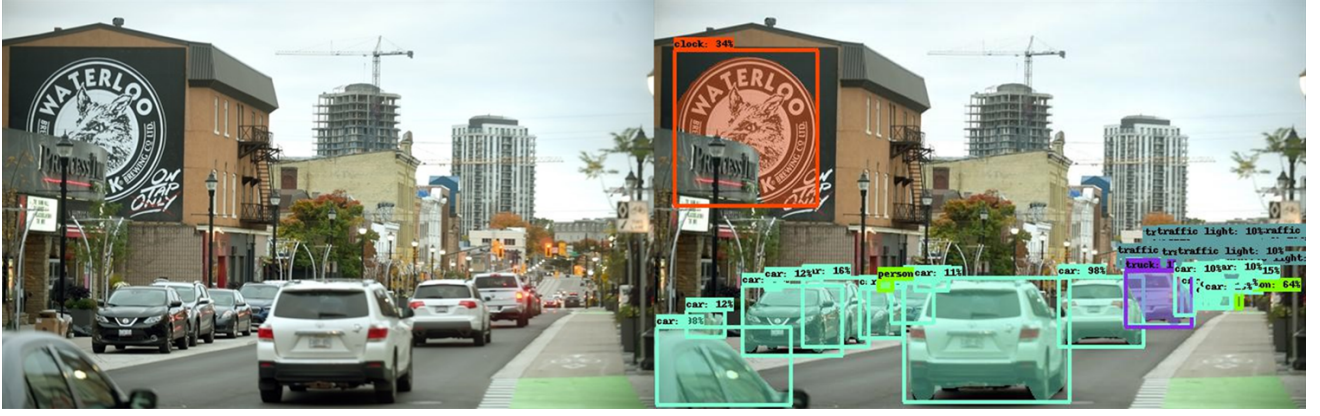


Fig. 1. Example of road-scene perception in Waterloo, Ontario, illustrating the role of dense visual understanding in autonomous-driving applications.

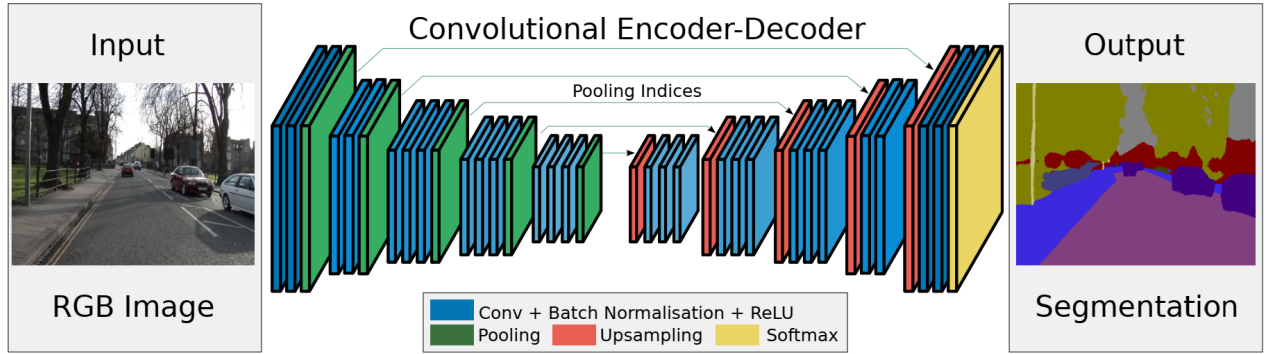


Fig. 2. SegNet encoder-decoder architecture [3].

B. LiDAR-Camera Fusion for Road Detection

Multimodal fusion aims to exploit the complementary strengths of different sensors. Caltagirone et al. proposed FCN-based camera-LiDAR fusion for road detection and compared early, late, and cross-fusion strategies [9]. Their results showed that fusion outperformed RGB-only processing on the KITTI road benchmark, demonstrating the value of combining appearance and geometric cues.

Lee and Park proposed a SegNet-based camera-LiDAR fusion approach using spherical-coordinate transformation [10]. Their method was designed to reduce spatial mismatch between camera and LiDAR representations while maintaining a computationally efficient encoder-decoder structure.

PLARD introduced progressive LiDAR adaptation to better align LiDAR data and features with the visual domain [4]. Instead of relying on a single fusion stage, PLARD progressively adapts and combines LiDAR and image features, achieving strong performance on KITTI.

These studies suggest that fusion location and feature alignment are important design choices. The present work therefore focuses on a controlled comparison between early fusion and late feature fusion using a common modified SegNet backbone.

III. METHODOLOGY

A. Base Network

The proposed road-segmentation network is based on a SegNet-style encoder-decoder architecture, shown in Fig. 3. The network is adapted for binary road segmentation and uses convolutional blocks with batch normalization, ReLU activation, and dropout. Transposed convolutions are used in the decoder to recover spatial resolution.

The ReLU activation is defined as

$$f(x) = \begin{cases} 0, & x < 0, \\ x, & x \geq 0. \end{cases} \quad (1)$$

The final binary output uses a sigmoid activation,

$$S(x) = \frac{1}{1 + e^{-x}}. \quad (2)$$

Dropout is used as a regularization mechanism to reduce overfitting [11]. The decoder uses transposed convolution to increase spatial resolution after the encoder has compressed the input representation [12].

B. Fusion Strategies

Two multimodal fusion strategies are evaluated, as illustrated in Fig. 4.

TABLE I
REPRESENTATIVE LIDAR-CAMERA ROAD-SEGMENTATION METHODS REPORTED IN THE LITERATURE.

Method	Fusion Strategy	Base Architecture	Dataset	Reported MaxF
Caltagirone et al. [9]	Early/Cross/Late	FCN	KITTI	96.03%
Lee and Park [10]	Early	SegNet	KITTI	94.29%
Chen et al. [4]	Progressive/Cross	PSPNet	KITTI	97.03%

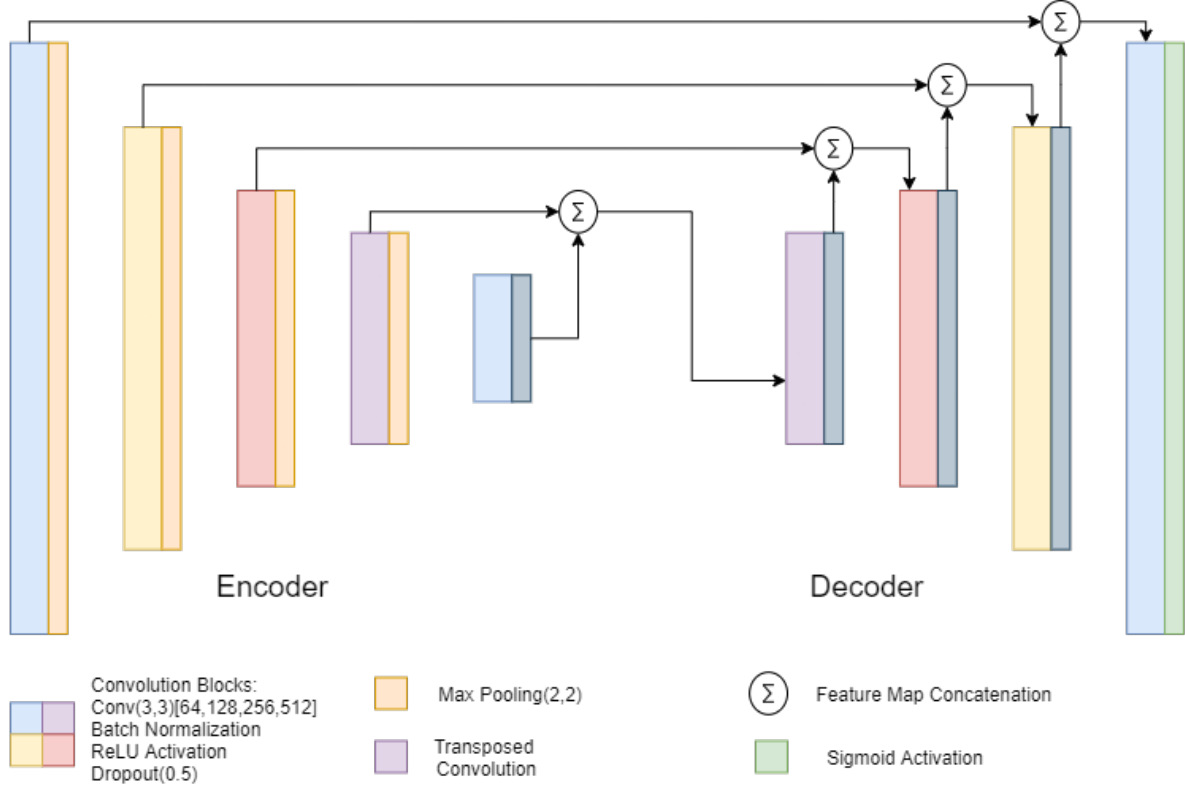


Fig. 3. Modified encoder-decoder architecture used in this work.

Early fusion: RGB and LiDAR-derived depth are aligned spatially and concatenated before entering the encoder. The network therefore learns a joint representation directly from low-level multimodal input.

Late feature fusion: RGB and LiDAR-derived depth are processed by separate encoder branches. Their encoded feature representations are concatenated before the decoder. This approach preserves modality-specific processing for a larger portion of the network at the cost of additional parameters and computation.

IV. DATA PROCESSING

A. KITTI Sensor Data

The KITTI dataset provides synchronized camera imagery, Velodyne HDL-64E LiDAR scans, and calibration parameters between the sensors [13]. Figure 5 shows a representative RGB frame and the corresponding LiDAR point cloud.

The experiments use camera 2 and the Velodyne LiDAR sensor. KITTI supplies the rigid-body calibration needed to transform LiDAR points into the camera frame.

B. LiDAR-to-Camera Projection

Let the homogeneous transformation from the Velodyne coordinate frame to the camera coordinate frame be

$$\mathbf{T}_{\text{velo}}^{\text{cam}} = \begin{bmatrix} \mathbf{R}_{\text{velo}}^{\text{cam}} & \mathbf{t}_{\text{velo}}^{\text{cam}} \\ \mathbf{0}^T & 1 \end{bmatrix}, \quad (3)$$

where

$$\mathbf{R}_{\text{velo}}^{\text{cam}} \in \mathbb{R}^{3 \times 3}, \quad \mathbf{t}_{\text{velo}}^{\text{cam}} \in \mathbb{R}^3. \quad (4)$$

A homogeneous LiDAR point $\mathbf{x}$ is projected into camera i as

$$\tilde{\mathbf{y}} = \mathbf{P}_{\text{rect}}^i \mathbf{R}_{\text{rect}}^0 \mathbf{T}_{\text{velo}}^{\text{cam}} \mathbf{x}, \quad (5)$$

followed by homogeneous normalization to obtain image coordinates.

Projected LiDAR samples that fall inside the camera field of view are retained. Pixels with no valid LiDAR correspondence

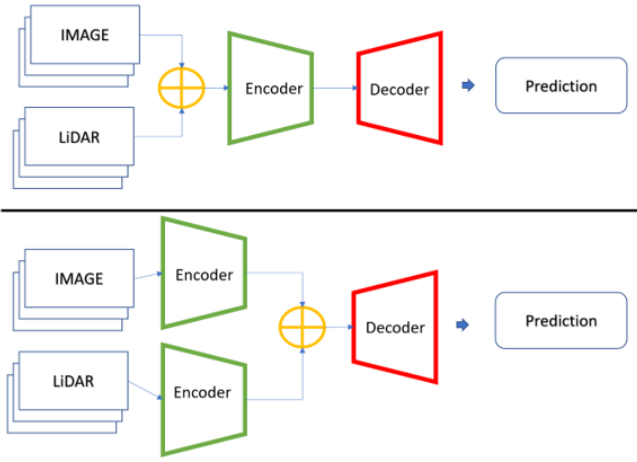


Fig. 4. Fusion strategies evaluated in this work. Top: early fusion at the input level. Bottom: late feature fusion after independent RGB and LiDAR encoding.

are assigned a default value. The resulting sparse representation is converted into an image-aligned depth map and resized to the network input resolution.

V. EXPERIMENTAL SETUP

A. Dataset Split

The reported experiments use KITTI road data together with the corresponding LiDAR scans and road labels. The project records 289 labeled road images and 96 held-out samples used for evaluation. The road scenes include urban marked, urban unmarked, and urban multi-lane cases.

B. Training Configuration

Training was performed in TensorFlow 2/Keras using a GPU-enabled Kaggle Notebook environment. Adam [14] was used with a learning rate of 10^{-4} , $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-7}$. All reported models were trained for 30 epochs with a batch size of 8.

Performance was summarized using MaxF, pixel accuracy, and Intersection over Union (IoU). The same training configuration was used across the RGB-only and fusion experiments unless otherwise noted.

VI. RESULTS AND DISCUSSION

A. Modified SegNet Baseline

The first experiment compares the baseline SegNet implementation with the modified network using RGB input only. The modified model improves the reported MaxF from 0.8142 to 0.9010, as summarized in Table II.

The improved MaxF indicates that the modified architecture is better suited to the binary road-segmentation task under the reported experimental configuration. This gain comes with a substantial increase in parameter count, so the improvement should be interpreted together with the higher model complexity.

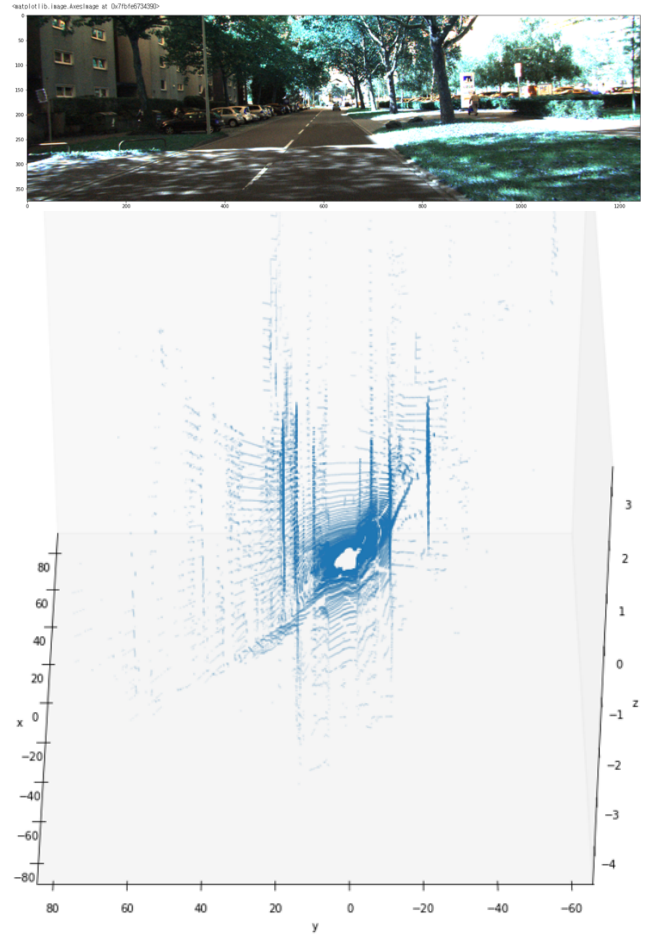


Fig. 5. Example KITTI RGB image and corresponding LiDAR point cloud.

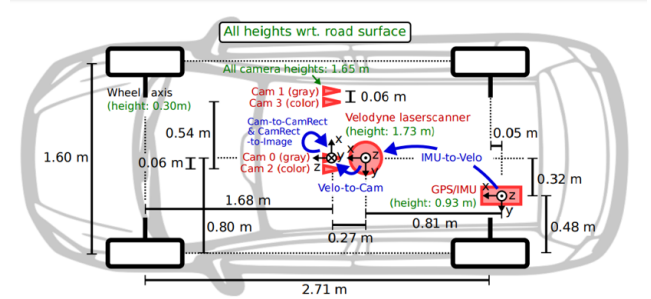


Fig. 6. KITTI sensor configuration showing the camera and Velodyne LiDAR locations [13].

B. LiDAR-Camera Fusion

The second experiment compares RGB-only segmentation with early and late LiDAR-camera fusion. Table III summarizes the reported results.

Early fusion increases MaxF from 0.9010 to 0.9153 while retaining the same reported training time as the RGB-only model. Late fusion produces the strongest result, increasing MaxF to 0.9364 and also achieving the highest reported accuracy and IoU. The improvement is accompanied by a longer

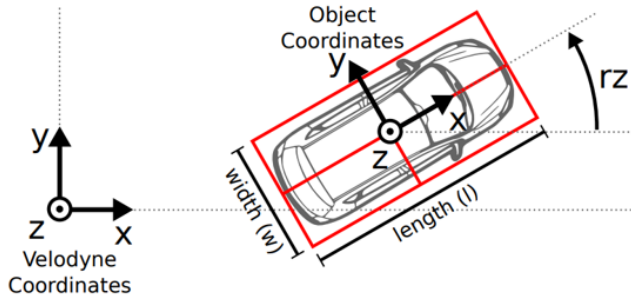


Fig. 7. KITTI Velodyne coordinate convention used for projection [13].

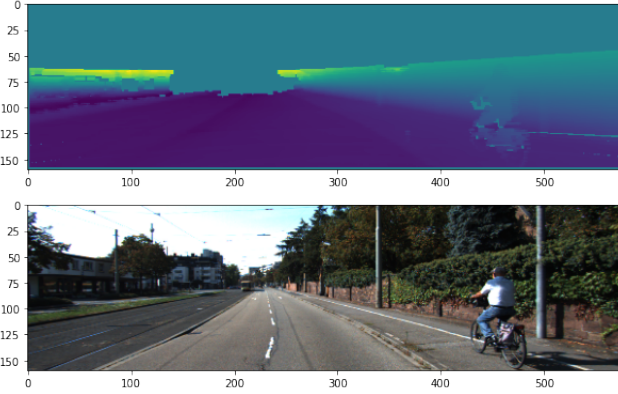


Fig. 8. LiDAR-derived depth representation and its corresponding RGB image after projection onto the camera plane.

training time of 584 s because the late-fusion architecture uses separate encoder branches for RGB and LiDAR inputs.

An important observation is that MaxF improves monotonically from RGB only to early fusion and then to late fusion, whereas IoU decreases slightly for early fusion before reaching its highest value under late fusion. This indicates that the benefit of early fusion is not uniform across all evaluation metrics. Late fusion provides the most consistent overall improvement, suggesting that maintaining modality-specific feature extraction before fusion is advantageous for this task.

VII. LIMITATIONS

The study has several limitations. First, the experiments were conducted on a relatively small dataset and in a cloud-based training environment, which constrained the extent of architecture and hyperparameter exploration. Second, the available KITTI road scenes do not cover the full range of adverse weather, illumination, and traffic conditions encountered in production autonomous-driving systems. Third, the experiments evaluate only RGB-only, early-fusion, and late-fusion configurations; cross-fusion and more advanced feature-adaptation strategies were not implemented. Finally, the reported results are specific to the experimental split and implementation used in this project and should not be inter-

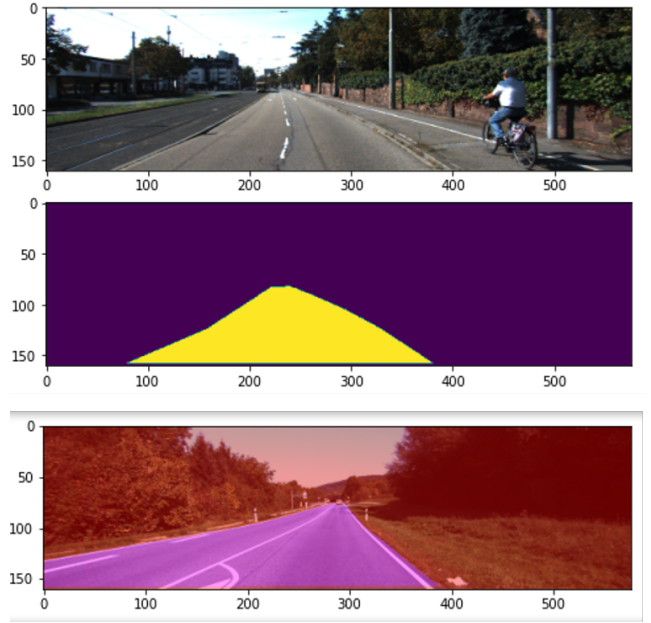


Fig. 9. Example KITTI road image, ground-truth road mask, and projected segmentation visualization.

TABLE II
BASELINE SEGNET AND MODIFIED SEGNET COMPARISON.

Metric	SegNet	Modified SegNet
Total parameters	10,191,489	22,735,297
Trainable parameters	10,185,601	22,725,569
Non-trainable parameters	5,888	9,728
MaxF	0.8142	0.9010

preted as a direct replacement for official KITTI benchmark submissions.

VIII. CONCLUSION

This work investigated LiDAR–camera fusion for binary road segmentation using a modified SegNet-style encoder–decoder network. KITTI LiDAR measurements were projected into the camera image plane to construct image-aligned depth representations, enabling direct comparison between RGB-only, early-fusion, and late-fusion configurations.

The modified RGB-only network improved the reported MaxF from 0.8142 for the baseline SegNet implementation to 0.9010. Adding LiDAR further increased MaxF to 0.9153 with early fusion and 0.9364 with late fusion. Late fusion also achieved the highest reported accuracy and IoU, although its two-branch encoder increased training time from 392 s to 584 s.

These results support the use of complementary geometric information for road segmentation and show that fusion location can significantly influence multimodal perception performance. Future work should evaluate the approach on more diverse road and weather conditions, investigate real-time architectures, and compare late fusion with cross-fusion or progressive feature-adaptation strategies.

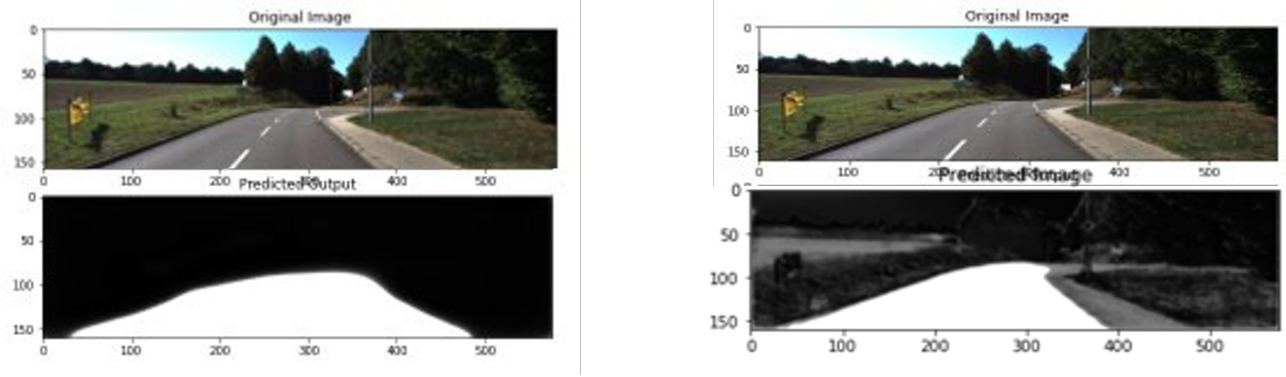


Fig. 10. Qualitative comparison between the baseline SegNet implementation and the modified RGB-only model.

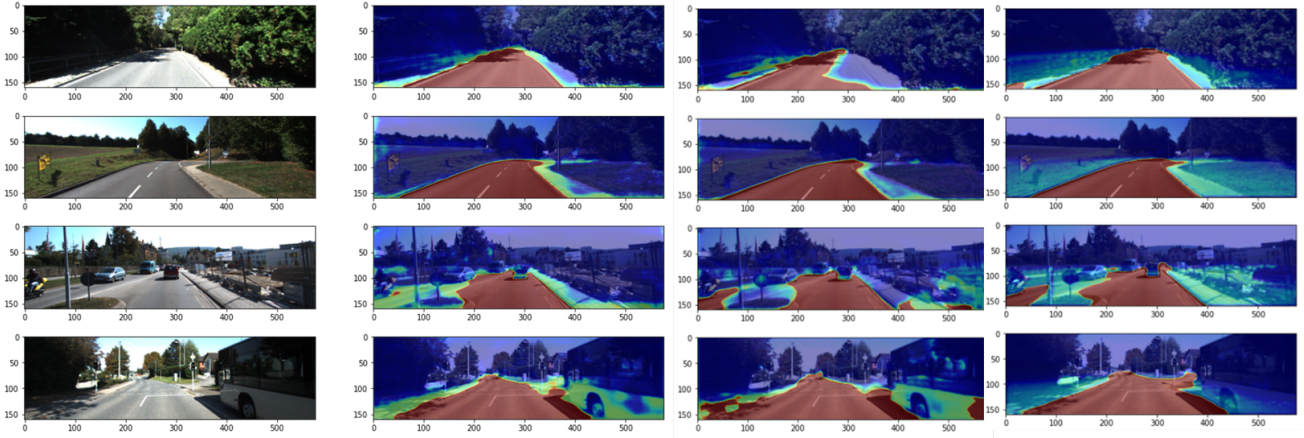


Fig. 11. Qualitative comparison of RGB-only, early-fusion, and late-fusion road segmentation. Columns show the input image and the corresponding segmentation results for each sensor configuration.

TABLE III
REPORTED PERFORMANCE OF RGB-ONLY AND MULTIMODAL FUSION CONFIGURATIONS.

Configuration	MaxF	Accuracy	IoU	Train Time [s]
RGB only	0.9010	0.7828	0.4755	392
Early fusion	0.9153	0.7875	0.4665	392
Late fusion	0.9364	0.7958	0.4898	584

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